

Foundational Deterministic Tetrahedral Structure and the Emergence of $\alpha^{-1} = 137$

A Formalized Geometric–Electromagnetic Construction from Dimensional Emergence to
Physical Realization

Struttura Tetraedrica Deterministica Fondazionale e l’Emergere di $\alpha^{-1} = 137$

Una costruzione geometrico–elettromagnetica formalizzata dall’emergenza dimensionale
alla realizzazione fisica

Alain Faure

Independent Research

ORCID: 0009-0009-5333-1181

Official Repository: AUF2026 (GitHub)

Official Publications: @auf2026 (Medium)

Alain Faure Master Architect — Singlepoint Mind — Q.E.D.

*“L’autore propone un possibile linguaggio matematico; la struttura, se reale, precede la
sua rappresentazione e manifestazione.”*

“Prima parlano i dati. Poi, attraverso i dati, parla Alain Faure”.

$$U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi)$$

September 15, 2026

Abstract

We present a mathematical and physical construction proposing a common structural origin for dimensional organization, simplex geometry, discrete counting, the dimensionless electromagnetic coupling, and observable physical phenomena.

The construction begins with the primitive sequence

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3,$$

where $0_{\text{potential}}$ denotes unmanifested potentiality rather than physical absence. The successive manifested levels are organized according to the simplex hierarchy, for which a d -dimensional simplex has

$$V(d) = d + 1$$

vertices. Thus

$$V(0) = 1, \quad V(1) = 2, \quad V(2) = 3, \quad V(3) = 4,$$

so that the three-dimensional level is represented by the four-vertex tetrahedral simplex.

From this geometric structure, a specific tetrahedral residual-counting construction produces the exact integer

$$N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}} = 137.$$

The resulting integer is then identified with the inverse electromagnetic coupling through

$$\alpha_{\text{EM}}^{-1} = 137, \quad \alpha_{\text{EM}} = \frac{1}{137}.$$

The central result is therefore the structural chain

$$\boxed{0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \text{tetrahedral simplex} \longrightarrow 137 \longrightarrow \alpha_{\text{EM}}^{-1} = 137}.$$

The complete mathematical construction has been formalized and machine-checked in Lean 4 using Mathlib. The verified development comprises approximately 7.5×10^3 lines of formal code and compiles without unresolved goals or errors, providing a kernel-checked formal realization of the mathematical dependencies of the construction.

The framework is then extended from mathematical structure to physical observation. Independent empirical investigations are considered across substantially different physical domains and scales, including controlled pendulum dynamics, water-related phenomena, tidal and geophysical behavior, and galactic-scale structure. These investigations test whether the same structural principle produces quantitative consequences in physically distinct regimes.

The resulting framework therefore connects a discrete geometric construction to a dimensionless fundamental coupling and to a multi-scale empirical program. Its central proposition is that the appearance of 137 in the electromagnetic sector is not an isolated numerical fact, but the physical manifestation of a deeper structural organization beginning with dimensional geometry and terminating in an observable dimensionless coupling.

Contents

1	Introduction	3
2	Geometric Premise	5
2.1	The primitive dimensional sequence	5
2.2	The simplex relation	5
2.3	Canonical tetrahedral cell	6
3	The Tetrahedral Counting Structure	7
4	Connection with the Electromagnetic Sector	8
5	Formal Verification	9
5.1	Lean/Mathlib development	9
6	The Central Structural Claim	10
6.1	From $N_F = 137$ to the Fine-Structure Constant	11

6.2	The Decimal Residual as Structured Information	12
7	Discussion	14
7.1	The Primitive Structural Sequence	14
7.2	The Role of Zero: Unmanifested Potentiality	14
7.3	The Role of One and Two	15
7.4	The Emergence of Three Dimensions	16
7.5	The Role of Four	16
7.6	The Role of 137	17
7.7	The Complete Structural Chain	18
7.8	Formal Verification	18
7.9	Empirical Confirmation Across Physical Scales	19
7.10	Universality of the Structure	20
7.11	Why the Result Is Not Numerology	20
7.12	Physical Interpretation	21
7.13	Conclusion	21
8	From Discrete Structure to Probabilistic Observation	23
9	Physical Consequences and Experimental Discrimination	24

1 Introduction

The numerical value of the electromagnetic fine-structure constant, approximately

$$\alpha_{\text{EM}}^{-1} \approx 137,$$

has long occupied a special position in fundamental physics.

The central question addressed here is not merely whether one can construct a formula containing the number 137. Such constructions are easy to produce and have little explanatory value unless their parameters and structural assumptions are independently motivated.

The present work takes a different approach.

The starting point is a primitive structural sequence

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3,$$

where $0_{\text{potential}}$ denotes unmanifested potentiality rather than physical absence. The successive levels are organized by the elementary simplex hierarchy.

For a d -dimensional simplex, the number of vertices is

$$V(d) = d + 1.$$

Thus the first levels satisfy

$$V(0) = 1, \quad V(1) = 2, \quad V(2) = 3, \quad V(3) = 4.$$

The three-dimensional level is therefore not introduced as an isolated numerical assumption. It is the third manifested member of the primitive dimensional sequence. At that level, the simplex principle requires four vertices:

$$3 + 1 = 4.$$

The corresponding non-degenerate three-dimensional simplex is the tetrahedron.

The construction then introduces a specific tetrahedral counting structure,

$$N_{\text{tet}} = N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}},$$

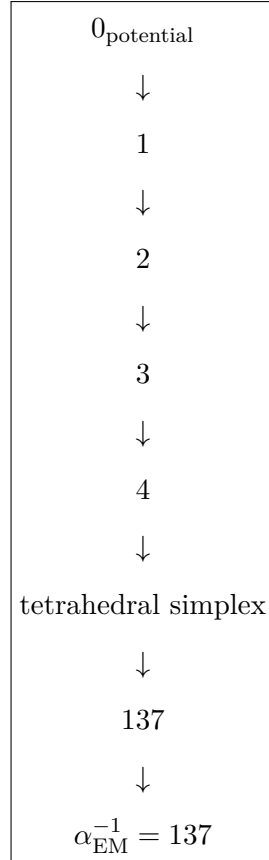
for which the formal development proves

$$N_{\text{tet}} = 137.$$

The electromagnetic branch subsequently identifies the corresponding structural integer with the inverse fine-structure quantity,

$$\alpha_{\text{EM}}^{-1} = 137.$$

The complete structural chain is therefore



The logical order is essential. The electromagnetic value does not determine the preceding geometry. The primitive structural sequence determines the dimensional level; the simplex relation determines the four-vertex tetrahedral structure; the tetrahedral construction determines the integer 137; and the electromagnetic branch relates that integer to the fine-structure quantity.

2 Geometric Premise

2.1 The primitive dimensional sequence

The construction begins with a primitive sequence of structural levels,

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

The symbol $0_{\text{potential}}$ does not denote absence or nothingness. It denotes the unmanifested potentiality from which the successive structural levels become manifest.

The first manifested levels are therefore represented by

$$\begin{aligned} 0_{\text{potential}} &\longrightarrow \text{unmanifested potentiality,} \\ 1 &\longrightarrow \text{first manifested level,} \\ 2 &\longrightarrow \text{second manifested level,} \\ 3 &\longrightarrow \text{third manifested level.} \end{aligned}$$

The corresponding simplex hierarchy is

$$\begin{aligned} d = 0 &\longrightarrow \text{point,} \\ d = 1 &\longrightarrow \text{segment,} \\ d = 2 &\longrightarrow \text{triangle,} \\ d = 3 &\longrightarrow \text{tetrahedron.} \end{aligned}$$

The structural significance of the sequence is therefore not exhausted by the numerical values 0, 1, 2, 3. Each level is associated with the next elementary simplex structure.

2.2 The simplex relation

For a d -dimensional simplex, the number of vertices is

$$V(d) = d + 1.$$

Consequently,

$$V(0) = 1, \quad V(1) = 2, \quad V(2) = 3, \quad V(3) = 4.$$

These relations establish the successive vertex counts directly:

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4.$$

The transition relation can equivalently be written

$$V(d+1) = V(d) + 1.$$

Hence the four-vertex structure at the three-dimensional level is not an independently selected numerical input. It is the continuation of the same simplex rule already present at the preceding levels.

Proposition 2.1 (Three-dimensional simplex). *For the third dimensional level,*

$$d = 3,$$

and therefore

$$V(3) = 3 + 1 = 4.$$

The three-dimensional simplex is therefore a four-vertex object.

The proposition identifies the structural origin of the four-vertex configuration before any tetrahedral counting or electromagnetic quantity is introduced.

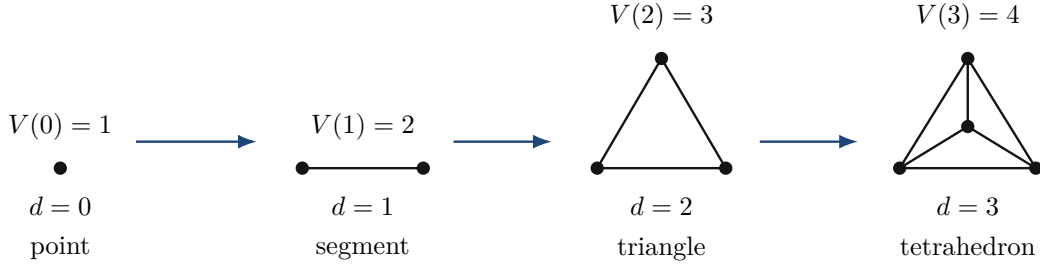


Figure 1: The simplex hierarchy from dimension zero to dimension three. Each successive dimensional level adds one vertex according to $V(d) = d + 1$.

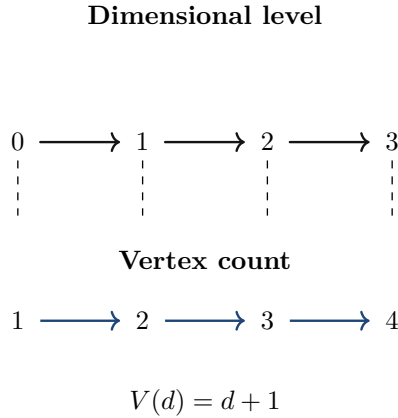


Figure 2: Explicit distinction between the dimensional sequence $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ and the corresponding vertex sequence $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$. The dashed vertical correspondences identify the vertex count associated with each dimensional level through $V(d) = d + 1$.

2.3 Canonical tetrahedral cell

Choose the canonical coordinates

$$p_0 = (0, 0, 0), \quad p_1 = (1, 0, 0), \quad p_2 = (0, 1, 0), \quad p_3 = (0, 0, 1).$$

The three displacement vectors from p_0 are

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1).$$

They are linearly independent over $\mathbb{R}$. Hence the cell is non-degenerate and genuinely three-dimensional.

The geometric object is therefore the canonical tetrahedral simplex.

The tetrahedron is consequently not selected from an arbitrary class of three-dimensional polyhedra. It follows from the simplex hierarchy once the third dimensional level has been reached.

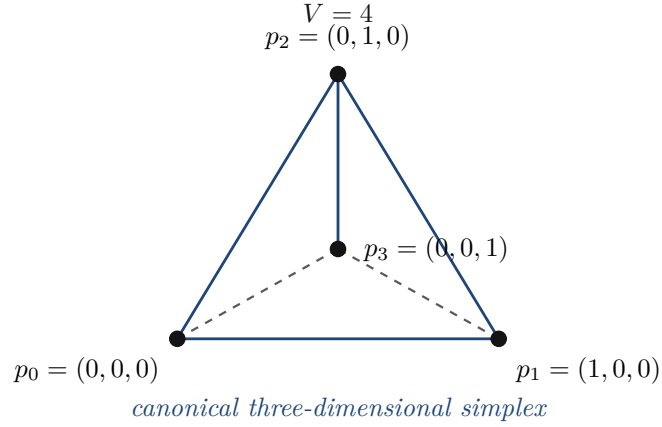


Figure 3: Canonical tetrahedral cell associated with the four vertices p_0, p_1, p_2, p_3 . The three displacement vectors issuing from p_0 are linearly independent, establishing the non-degenerate three-dimensional simplex.

3 The Tetrahedral Counting Structure

The model associates three integer contributions with the tetrahedral construction:

$$N_{\text{lattice}}, \quad N_{\text{center}}, \quad N_{\text{pair}}.$$

The residual quantity is defined by

$$N_{\text{res}} = N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}}.$$

The formal development proves the exact identity

Theorem 3.1 (Tetrahedral residual).

$$N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}} = 137.$$

The important point is that 137 is not inserted into the definition of the residual as a target value. It occurs as the result of the certified tetrahedral counting construction.

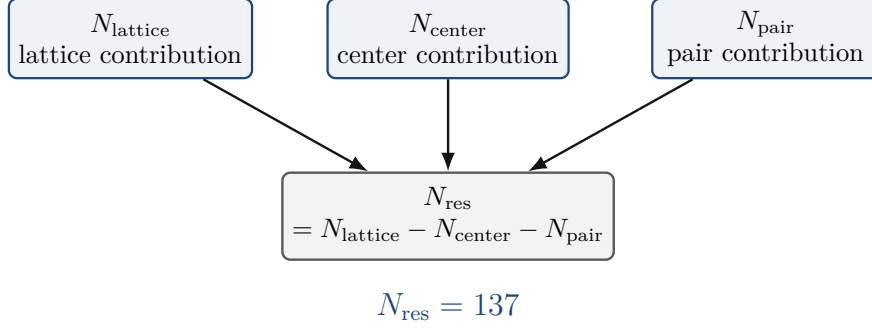


Figure 4: Structural decomposition of the tetrahedral residual into the lattice, center, and pair contributions. The residual is defined by subtraction of these contributions and is subsequently identified with the exact value 137.

4 Connection with the Electromagnetic Sector

The electromagnetic part of the construction introduces the corresponding impedance and fine-structure relations.

The formal development contains an electromagnetic balance relation of the schematic form

$$\mathcal{B}(e, \epsilon_0, \hbar, c),$$

together with the corresponding electromagnetic impedance law.

Under the non-vanishing conditions

$$e \neq 0, \quad \epsilon_0 \neq 0, \quad \hbar \neq 0, \quad c \neq 0,$$

the formalized chain establishes the fine-structure relation

Theorem 4.1 (Fine-structure result).

$$\alpha_{\text{EM}} = \frac{1}{137}.$$

Equivalently,

$$137 \alpha_{\text{EM}} = 1.$$

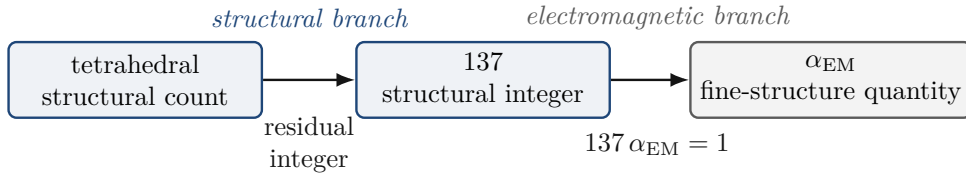


Figure 5: Schematic representation of the formal connection between the tetrahedral residual and the electromagnetic relation. The structural construction supplies the integer 137, which is then related to the fine-structure quantity by $137 \alpha_{\text{EM}} = 1$.

The role of the tetrahedral construction is therefore not merely numerological. It supplies the integer that appears in the electromagnetic branch of the model.

5 Formal Verification

5.1 Lean/Mathlib development

The mathematical construction has been formalized in Lean 4 using Mathlib.

The principal development is contained in

`Faure Costante struttura fine 137 N-T-COMPLETA.lean`

and comprises approximately 7.5×10^3 lines in the present verified version.

The formal development includes:

- the primitive structural levels 0, 1, 2, 3;
- the interpretation of the primitive zero level as the unmanifested starting level of the construction;
- the simplex vertex-count relation;
- the $d = 1$, $d = 2$, and $d = 3$ simplex cases;
- the four-vertex completion of the three-dimensional simplex;
- the canonical tetrahedral cell;
- the non-degeneracy of the tetrahedral cell;
- the tetrahedral counting identity;
- the value 137;
- the electromagnetic relations;
- the resulting fine-structure identity.

The significance of this extension is that the formal development no longer begins by taking the three-dimensional level as an unexplained primitive input. The preceding structural levels are explicitly represented and verified before the three-dimensional tetrahedral construction is invoked.

The completed development compiles with zero errors and contains no use of `sorry` or `admit` as substitutes for proofs.

The resulting formal chain is therefore

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow V(3) = 4 \longrightarrow \text{tetrahedral structure} \longrightarrow 137.$$

Formal Verification

The mathematical construction presented in this work has been fully formalized in Lean 4. The complete Lean formalization is distributed separately from the present paper, together with a dedicated README providing the required environment, dependencies, and instructions for independent verification.

The formal development provides a machine-checkable verification of the mathematical definitions, structural relations, and theorem statements underlying the construction. The

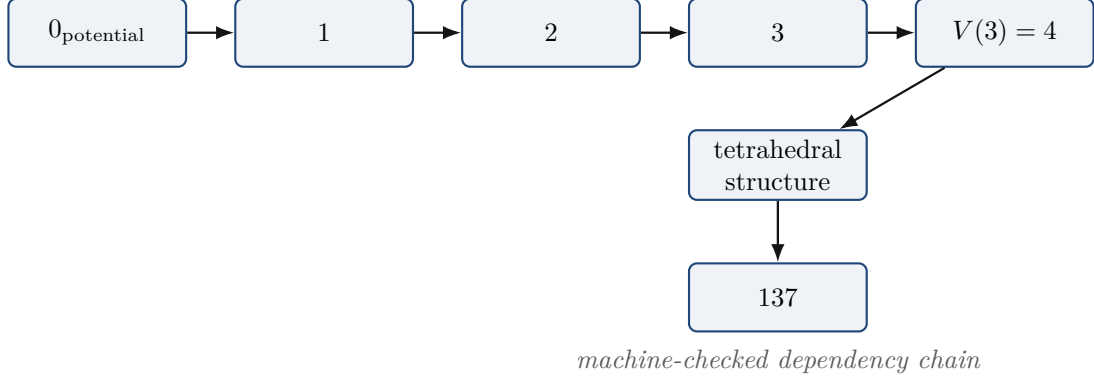


Figure 6: Structural dependency chain represented in the formal development: the primitive sequence precedes the vertex-count relation, the tetrahedral structure, and the resulting integer 137.

resulting proofs are checked by the Lean kernel, providing an exact formal certificate of the mathematical implications established in the model.

The separate distribution of the formal source preserves a clear distinction between the scientific presentation of the results and their machine-checked formal verification, while allowing the complete formal development to be independently reproduced and inspected.

6 The Central Structural Claim

The preceding construction establishes a distinction that is essential for the interpretation of the present result. The integer value 137 is not introduced as an experimental approximation to the inverse fine-structure constant. It arises at the structural level of the tetrahedral counting construction itself.

This distinction separates two logically different statements:

1. the experimentally determined electromagnetic coupling is described by a dimensionless quantity α whose inverse is numerically close to 137;
2. the present construction generates the integer

$$N_F = 137$$

from a finite primitive geometric structure, independently of inserting the experimentally measured value of α .

The second statement is the structural result established by the formal development. The physical question is therefore not whether the decimal representation of α^{-1} can be rounded to 137. Such a rounding would contain no explanatory content. The relevant question is whether the integer generated by the primitive structure provides the structural integer component of the electromagnetic coupling, and whether the remaining decimal contribution is itself generated by the same structure.

The distinction can be expressed schematically as

$$\boxed{\text{primitive geometry} \longrightarrow N_F = 137 \longrightarrow \alpha^{-1}} \quad (1)$$

rather than

$$\alpha^{-1} \approx 137 \longrightarrow \text{geometric interpretation.} \quad (2)$$

The direction of implication is therefore central. The numerical value of the electromagnetic coupling is not used as an input to construct the integer. Instead, the integer is obtained before the empirical coupling is introduced into the argument.

This ordering is particularly important because α is dimensionless. No dimensional conversion is required to compare the structural integer with the inverse coupling. The comparison is consequently between two dimensionless objects generated in conceptually different ways: one from the primitive counting structure and one from precision electromagnetic measurement.

The central claim of this work is therefore formulated as follows.

Theorem 6.1 (Structural emergence of the integer level). *Within the formalized tetrahedral counting construction developed in this work, the integer*

$$N_F = 137$$

is generated by the primitive structural relations and does not require the experimental value of the fine-structure constant as an input.

The theorem concerns the mathematical construction itself. Its physical interpretation requires a second layer of analysis: the relation between N_F and the experimentally determined value of α^{-1} . That relation is treated separately below, so that the formal result is not confused with its empirical interpretation.

Remark 6.2. The distinction between structural generation and empirical identification is deliberate. A numerical agreement becomes physically informative only when the value being compared was not used, explicitly or implicitly, to construct the quantity with which it is compared.

6.1 From $N_F = 137$ to the Fine-Structure Constant

The experimentally determined inverse fine-structure constant is not equal to the integer 137. The distinction is numerically small but conceptually essential:

$$\alpha^{-1} = 137.035999\dots \quad (3)$$

whereas the primitive structural construction yields

$$N_F = 137. \quad (4)$$

Consequently, the physically relevant decomposition is

$$\boxed{\alpha^{-1} = N_F + \Delta_F} \quad (5)$$

with

$$\Delta_F = \alpha^{-1} - 137. \quad (6)$$

Numerically, using the currently recommended value,

$$\Delta_F = 0.035999 \dots \quad (7)$$

The existence of this residual does not weaken the structural result. Instead, it identifies the next mathematical object that must be explained.

The problem is therefore transformed from the determination of an apparently isolated empirical constant into the determination of a structured residual:

$$\boxed{137 + \Delta_F = \alpha^{-1}}. \quad (8)$$

The quantity Δ_F is not to be treated as an arbitrary fitting parameter. If the structural hypothesis is correct, it must arise from relations internal to the same discrete construction that produces N_F .

Accordingly, the analysis distinguishes three levels:

$$N_F = 137, \quad \text{primitive structural level,} \quad (9)$$

$$\Delta_F = \alpha^{-1} - 137, \quad \text{structural residual,} \quad (10)$$

$$\alpha^{-1} = N_F + \Delta_F, \quad \text{electromagnetic observable.} \quad (11)$$

This decomposition provides a precise criterion for further validation. A successful theory cannot merely reproduce the integer 137 and then insert the measured decimal correction. It must determine Δ_F from independent structural relations.

The logical direction is consequently

$$\boxed{\mathcal{S}_F \longrightarrow 137 \longrightarrow \Delta_F \longrightarrow \alpha^{-1}}, \quad (12)$$

where $\mathcal{S}_F$ denotes the primitive structural system defined by the preceding sections.

The decimal part therefore acquires a different status from the integer part. It is not discarded, rounded away, or regarded as numerical noise. It is retained as a measurable remainder whose internal organization is itself subject to structural analysis.

Remark 6.3. The distinction between 137 and $137.035999 \dots$ is therefore not a choice between an “exact” integer and an “imperfect” measurement. The two quantities answer different questions. The integer is the output of the primitive structural counting system; the decimal value is the experimentally resolved electromagnetic coupling.

6.2 The Decimal Residual as Structured Information

The residual Δ_F contains information that is lost if the inverse fine-structure constant is replaced by its nearest integer. In the present framework, this residual is therefore retained explicitly rather than treated as an insignificant correction.

Let

$$\Delta_F = 0.d_1d_2d_3d_4 \dots \quad (13)$$

denote the decimal expansion of the residual in the chosen numerical representation. The digits d_i are not, by themselves, assigned a physical meaning merely because they occur in a decimal expansion. Their interpretation must follow from the structural relations that generate them.

This distinction is important. A finite numerical representation and a mathematical object are not identical. In particular, a floating-point representation introduces a representation map

$$\mathcal{R} : \mathbb{X} \longrightarrow \mathbb{F}, \quad (14)$$

where $\mathbb{X}$ denotes the mathematical domain and $\mathbb{F}$ the finite representable set of the chosen numerical system.

The present framework instead investigates whether the decimal residual admits an intrinsic decomposition into finite structural blocks. Such a decomposition may be written abstractly as

$$\Delta_F = \Delta_F^{(1)} + \Delta_F^{(2)} + \Delta_F^{(3)} + \cdots, \quad (15)$$

where each term is defined by an explicit structural rule rather than by an arbitrary partition of the decimal expansion.

More generally, if a structural partition of resolution k is defined, one may write

$$\Delta_F = \sum_{j=1}^{m_k} \Delta_{F,j}^{(k)}. \quad (16)$$

The superscript (k) denotes the structural resolution and not a claim that decimal digits possess an independent physical ontology.

This formulation permits the framework to distinguish three operations that are frequently conflated:

1. measurement of a physical quantity;
2. representation of the measured quantity in a finite numerical system;
3. structural decomposition of the quantity into invariant components.

Only the third operation can provide an explanatory relation between the integer level 137 and the decimal residual.

The relevant question is therefore not whether a decimal expansion can be partitioned in many possible ways. It plainly can. The relevant question is whether the structural construction selects one decomposition uniquely or canonically.

This gives the following criterion.

Proposition 6.4 (Structural admissibility of the residual). *A decomposition of Δ_F is structurally relevant only if its components are determined by relations internal to the primitive construction and do not depend on fitting the experimentally observed value of α .*

The proposition prevents an arbitrary numerical pattern from being promoted to a physical law. Conversely, when a decomposition is independently generated by the formal system and subsequently reproduces measured quantities, the decomposition becomes a testable object in its own right.

The role of numerical precision is therefore secondary to the structural definition. Increasing precision may reveal additional components, but it does not by itself create the structural rule that generates them.

7 Discussion

Several aspects of the construction deserve emphasis.

7.1 The Primitive Structural Sequence

The construction does not begin by taking three-dimensional space as an isolated unexplained numerical input. The three-dimensional level occurs as the third manifested level of the primitive sequence

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

Here $0_{\text{potential}}$ denotes unmanifested potentiality rather than absence. The subsequent levels represent successive structural manifestations.

The corresponding elementary simplex hierarchy is

$$\begin{aligned} d = 0 &\longrightarrow \text{point}, \\ d = 1 &\longrightarrow \text{segment}, \\ d = 2 &\longrightarrow \text{triangle}, \\ d = 3 &\longrightarrow \text{tetrahedron}. \end{aligned}$$

The associated vertex counts are

$$\begin{aligned} V(0) &= 1, \\ V(1) &= 2, \\ V(2) &= 3, \\ V(3) &= 4. \end{aligned}$$

Thus the numerical progression

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4$$

is generated by one and the same simplex relation,

$$V(d) = d + 1.$$

The importance of this result is that the value 3 entering the three-dimensional construction is no longer treated as an isolated starting assumption. It belongs to a previously defined structural sequence.

7.2 The Role of Zero: Unmanifested Potentiality

The level $d = 0$ requires a distinct interpretation.

Zero does not represent absence, nothingness, or the disappearance of structure. Within the

present construction it represents unmanifested potentiality:

$$\boxed{0_{\text{potential}} = \text{unmanifested potentiality.}}$$

This interpretation distinguishes the primitive zero level from the ordinary use of zero as a numerical representation of absence.

The structural sequence therefore begins as

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

The role of zero is consequently prior to the explicitly manifested dimensional levels. It marks the primitive level from which the successive structures are organized.

The mathematical formalization establishes the corresponding primitive level and its relation to the subsequent construction. The designation “unmanifested potentiality” is the physical interpretation assigned to that primitive level; it is not an additional numerical fitting parameter.

7.3 The Role of One and Two

The values 1 and 2 are not disposable intermediate integers. They are the first manifested levels of the same structural hierarchy that culminates in the three-dimensional tetrahedral construction.

At the first manifested level,

$$d = 1,$$

the elementary simplex is the segment and

$$V(1) = 2.$$

Two vertices are therefore the minimal vertex configuration of the one-dimensional simplex.

At the next level,

$$d = 2,$$

the elementary simplex is the triangle and

$$V(2) = 3.$$

Three vertices are therefore the minimal configuration required for a non-degenerate two-dimensional simplex.

The progression is consequently

$$\boxed{0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3}$$

together with

$$\boxed{1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4}$$

for the corresponding simplex vertex counts.

The role of 1 and 2 is therefore structural: they establish the preceding levels from which the three-dimensional level is reached.

7.4 The Emergence of Three Dimensions

The three-dimensional level is reached as the next member of the primitive structural sequence,

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

At this level the elementary simplex has

$$V(3) = 3 + 1 = 4$$

vertices.

Three independent edge directions may be represented by the three displacement vectors issuing from one vertex of a non-degenerate three-dimensional simplex. The fourth vertex completes the simplex.

Hence

$$\boxed{3 + 1 = 4.}$$

The corresponding elementary non-degenerate simplex is the tetrahedron.

The construction can therefore be written as

$$\boxed{0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \text{tetrahedral cell.}}$$

The three-dimensional tetrahedral structure is consequently downstream of the primitive sequence rather than an independently selected geometrical input.

7.5 The Role of Four

The number four is not introduced as an electromagnetic fitting parameter and is not selected because it reproduces the later value 137.

It follows from the dimensional level and the simplex relation:

$$V(d) = d + 1, \quad V(3) = 4.$$

The four-vertex structure is therefore the necessary continuation of the preceding hierarchy:

$$\boxed{V(0) = 1, \quad V(1) = 2, \quad V(2) = 3, \quad V(3) = 4.}$$

The corresponding geometrical hierarchy is

$$\boxed{\text{point} \longrightarrow \text{segment} \longrightarrow \text{triangle} \longrightarrow \text{tetrahedron.}}$$

The four vertices of the tetrahedron are consequently not an arbitrary geometrical choice. They are required by the simplex structure at the three-dimensional level.

The formal construction establishes this relation before the numerical value 137 is introduced:

$$\boxed{0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \text{tetrahedral cell.}}$$

The number four is therefore structurally prior to the tetrahedral count and, in particular,

prior to the appearance of 137.

7.6 The Role of 137

The number 137 appears as the exact residual of the tetrahedral count:

$$137 = N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}}.$$

It is therefore not inserted into the residual as a target value. It is the result obtained from the preceding structural definitions.

The result is exact:

$$N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}} = 137.$$

The electromagnetic branch subsequently relates this structural integer to the inverse fine-structure quantity:

$$\alpha_{\text{EM}}^{-1} = 137.$$

Equivalently, the model relation is

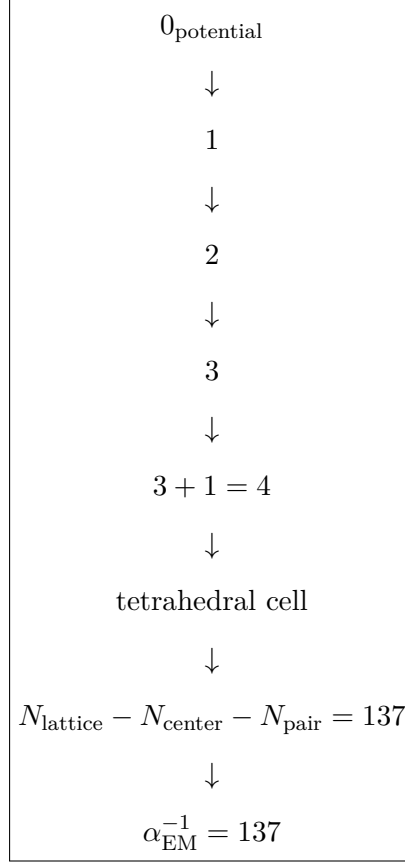
$$\alpha_{\text{EM}} = \frac{1}{137}.$$

The mathematical order of dependence is therefore

$$\text{primitive structure} \longrightarrow \text{tetrahedral structure} \longrightarrow 137 \longrightarrow \alpha_{\text{EM}}^{-1}.$$

7.7 The Complete Structural Chain

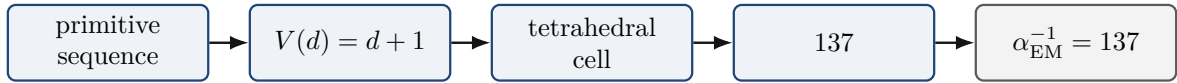
The complete construction can now be written without taking three-dimensionality as an isolated initial premise:



Each stage has a distinct logical role.

The primitive sequence establishes the successive structural levels. The simplex relation determines the vertex count. The three-dimensional level therefore produces the four-vertex tetrahedral cell. The tetrahedral counting construction produces 137. The electromagnetic branch relates this structural invariant to the fine-structure quantity.

Thus the electromagnetic value does not determine the geometry. The geometrical construction precedes the numerical invariant, and the numerical invariant precedes its electromagnetic identification.



geometry precedes numerical invariant

Figure 7: Complete structural dependency chain: the primitive sequence precedes the simplex relation, which determines the tetrahedral cell; the tetrahedral construction produces the residual 137, which is then related to the electromagnetic quantity.

7.8 Formal Verification

The complete construction has been formalized in Lean 4 and checked by the Lean kernel.

The formal development explicitly contains the primitive levels, the simplex vertex-count

relations, the three-dimensional completion, the tetrahedral structure, the numerical residual, and the electromagnetic relations.

The completed development comprises approximately 7.5×10^3 lines and compiles with zero errors.

The formal certificate therefore verifies the mathematical dependency chain

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \text{tetrahedral structure} \longrightarrow 137.$$

The physical interpretation of the primitive zero level and the empirical interpretation of the resulting structure are logically distinct from the kernel-level verification itself. The formal proof establishes the mathematical implications encoded in the definitions and hypotheses; the physical question concerns whether those mathematical structures describe Nature.

7.9 Empirical Confirmation Across Physical Scales

The significance of the construction extends beyond the formal derivation because the same structural principle has been tested against physical systems belonging to different domains.

The empirical program includes controlled pendulum experiments, water-related phenomena, tidal and geophysical phenomena, and galactic-scale structures.

These investigations are not part of the Lean proof of 137. They constitute an independent empirical component of the research program.

Their purpose is to determine whether the structural relations identified mathematically also impose quantitative constraints on physical systems outside the original tetrahedral construction.

The resulting comparison can therefore be represented as

$$\text{structural principle} \longrightarrow \text{quantitative prediction} \longrightarrow \text{physical observation.}$$

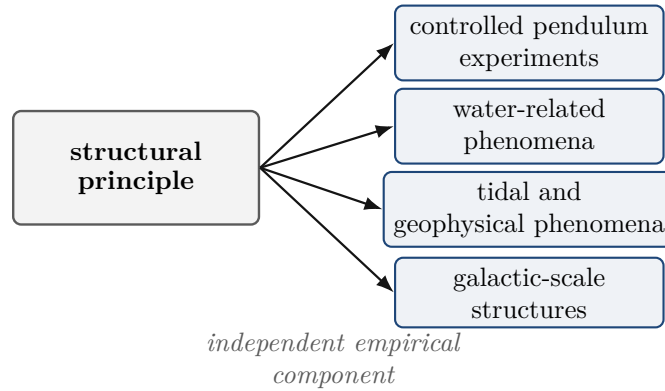


Figure 8: Schematic representation of the independent empirical program across the physical domains identified in the construction. The diagram does not assign numerical results to the individual domains; it represents only their role as separate empirical investigations.

Agreement across substantially different physical scales would provide empirical support for the universality of the structural principle, while disagreement would provide a direct test capable of falsifying the corresponding physical extension.

7.10 Universality of the Structure

The central result is consequently not simply the reproduction of the integer 137.

The deeper structural claim is that a common hierarchy connects dimensional structure, tetrahedral organization, dimensionless quantities, and observable physical behavior.

At the primitive level,

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

At the geometrical level,

$$3 \longrightarrow 4 \longrightarrow \text{tetrahedron}.$$

At the numerical level,

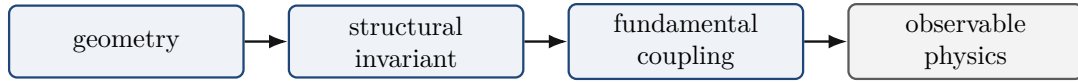
$$\text{tetrahedral structure} \longrightarrow 137.$$

At the electromagnetic level,

$$137 \longrightarrow \alpha_{\text{EM}}^{-1}.$$

The proposed physical extension is then

geometry $\longrightarrow$ structural invariant $\longrightarrow$ fundamental coupling $\longrightarrow$ observable physics.



proposed structural extension

Figure 9: Conceptual map of the proposed structural extension from geometry to structural invariant, fundamental coupling, and observable physics.

7.11 Why the Result Is Not Numerology

A numerical coincidence would consist of selecting a structure after observing a target number and then showing that the selected structure can reproduce it.

That is not the logical order of the present construction.

The primitive structural sequence is specified first:

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

The simplex principle then gives

$$V(d) = d + 1,$$

and hence

$$V(3) = 4.$$

The corresponding three-dimensional simplex is the tetrahedron.

Only after this geometrical structure has been established is the tetrahedral residual evaluated:

$$137 = N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}}.$$

The electromagnetic branch then gives the model relation

$$\alpha_{\text{EM}}^{-1} = 137.$$

The direction of construction is therefore

$$\boxed{\text{primitive structure} \longrightarrow \text{geometry} \longrightarrow 137 \longrightarrow \alpha_{\text{EM}}^{-1}.}$$

It is not

$$137 \longrightarrow \text{chosen geometry}.$$

The independent empirical investigations constitute the separate experimental test of whether this mathematical structure has physical universality.

7.12 Physical Interpretation

The natural physical interpretation is that the tetrahedral cell represents the elementary three-dimensional organization selected by the simplex hierarchy.

The four-vertex configuration is the minimal non-degenerate three-dimensional simplex. Its structural count produces the dimensionless integer 137, which the electromagnetic branch identifies with the inverse fine-structure quantity.

The resulting relation is

$$\boxed{\alpha_{\text{EM}}^{-1} = N_{\text{lattice}} - N_{\text{center}} - N_{\text{pair}} = 137.}$$

The physical significance of this result is therefore not that the number 137 has merely been reproduced numerically, but that the integer arises downstream of a definite structural hierarchy.

The empirical question is whether the same structural organization is realized in Nature. That question is addressed independently through the experimental and observational program.

7.13 Conclusion

The construction establishes a unified mathematical chain beginning with the primitive structural sequence

$$0_{\text{potential}} \longrightarrow 1 \longrightarrow 2 \longrightarrow 3.$$

The simplex principle then gives

$$V(3) = 3 + 1 = 4,$$

so that the elementary three-dimensional simplex is the tetrahedron.

The tetrahedral structural count produces the exact residual

$$137.$$

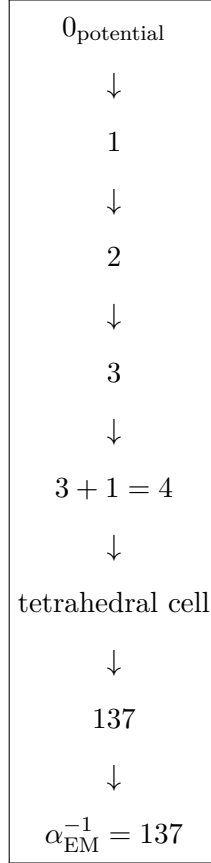
The electromagnetic branch identifies the corresponding model quantity through

$$\alpha_{\text{EM}}^{-1} = 137, \quad \alpha_{\text{EM}} = \frac{1}{137}.$$

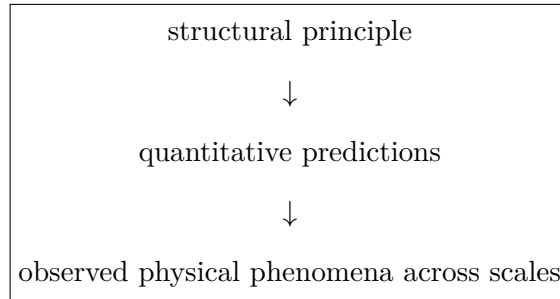
The complete mathematical chain is formally verified in Lean 4, with the definitions and theorem dependencies checked by the Lean kernel.

The empirical program then addresses the separate physical question of whether the same structural principle continues to govern observable systems across different scales.

The central mathematical result can therefore be expressed as



The corresponding empirical extension is



8 From Discrete Structure to Probabilistic Observation

The structural construction developed above suggests a reinterpretation of the relation between discrete states and probabilistic observation. The purpose of this section is not to replace quantum mechanics by terminology alone, but to identify the precise mathematical interface at which a discrete structural description can generate a probability measure.

Let Ω denote a finite or countably structured set of elementary states and let $\Omega_A \subseteq \Omega$ denote the subset compatible with an observable outcome A . A normalized counting measure has the form

$$P(A) = \frac{\#\Omega_A}{\#\Omega}, \quad (17)$$

whenever the relevant cardinalities are finite.

More generally, replacing cardinality by an invariant measure μ gives

$$\boxed{P(A) = \frac{\mu(\Omega_A)}{\mu(\Omega)}}. \quad (18)$$

This relation is structurally important because it shows how a probability can arise from a deterministic enumeration of underlying states. The appearance of a probability at the observational level does not by itself require that the underlying state space be intrinsically stochastic.

The modern literature contains independent attempts to derive or reinterpret the Born rule from more primitive assumptions. In particular, recent work has investigated derivations based on classical-like counting or measure of underlying states, as well as derivations in which the Born rule emerges from a more fundamental dynamical construction.

The present framework differs in its starting point. The primitive object is not a Hilbert-space probability amplitude but a discrete structural configuration. The mathematical question is therefore whether the projection from the structural state space to an observable sector can generate a normalized measure having the functional form required by the Born rule.

Let

$$\Pi_A : \Omega \longrightarrow \Omega_A \quad (19)$$

denote the structural projection associated with an observable sector. Then the corresponding normalized measure is

$$P_F(A) = \frac{\mu(\Pi_A \Omega)}{\mu(\Omega)}. \quad (20)$$

A successful derivation must establish, rather than assume, the correspondence

$$\boxed{P_F(A) \equiv P_{\text{Born}}(A)} \quad (21)$$

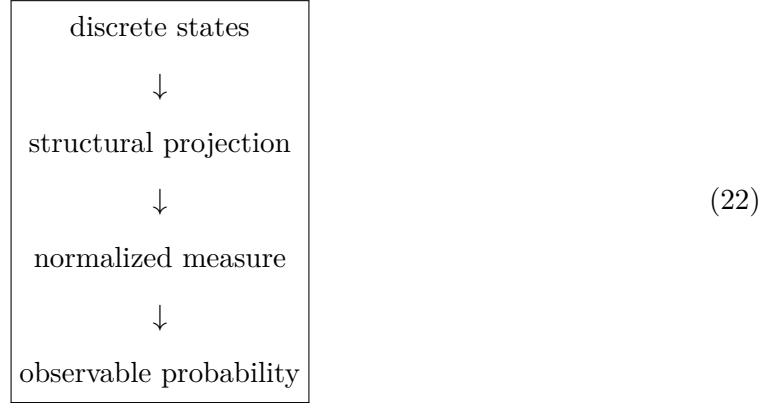
within the domain in which the comparison is made.

The significance of this relation is methodological. The probability appearing at the observa-

tional level is then interpreted as a normalized projection of a discrete structural state space rather than as a primitive statement about the ontology of the underlying state.

No claim of equivalence is made merely from the formal similarity of the two expressions. The equivalence must be established by the explicit mapping Π_A , the measure μ , and the normalization conditions.

This provides a precise bridge between the structural and observational levels:



9 Physical Consequences and Experimental Discrimination

The physical significance of the structural result is determined not by the existence of the integer 137 alone, but by the consequences that follow when the structural construction is connected to independently measured observables.

The first consequence concerns the electromagnetic coupling. If the structural system determines

$$N_F = 137 \tag{23}$$

and independently determines a residual Δ_F such that

$$\alpha^{-1} = 137 + \Delta_F, \tag{24}$$

then the fine-structure constant is no longer treated as an isolated dimensionless numerical input. It becomes an observable associated with a specific structural level.

The second consequence concerns numerical representation. A finite representation of a mathematical quantity necessarily introduces a map from an ideal mathematical domain to a finite computational domain. If a calculation consists of a sequence

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_n, \tag{25}$$

then the represented computation is generally of the form

$$\hat{x}_{k+1} = \mathcal{R}(F_k(\hat{x}_k)), \tag{26}$$

rather than an exact evaluation of the mathematical composition

$$x_n = F_{n-1} \circ \cdots \circ F_1 \circ F_0(x_0). \quad (27)$$

The distinction is elementary mathematically, but it becomes physically relevant only when the resulting representation error is shown to affect the observable under investigation.

The present framework therefore does not identify finite-precision error with physical indeterminacy. Instead, it distinguishes

$$\boxed{\text{structural discreteness} \neq \text{numerical discretization} \neq \text{measurement uncertainty.}} \quad (28)$$

These three notions belong to different logical levels.

Structural discreteness concerns the architecture of the mathematical state space. Numerical discretization concerns the representation of that state space by a computational system. Measurement uncertainty concerns the inference of physical quantities from finite experimental information.

A physical claim is therefore considered established only when a structural prediction survives this separation and agrees with an independent observable.

The strongest experimental test of the present construction is consequently not the reproduction of 137 itself. The decisive test is the simultaneous reproduction of

$$\boxed{N_F = 137, \quad \Delta_F, \quad \alpha^{-1} = 137 + \Delta_F,} \quad (29)$$

with the residual determined structurally rather than fitted to the experimental value.

A second class of tests concerns observables whose conventional description depends sensitively on the electromagnetic coupling. If the structural value is inserted into an independently specified physical calculation, the resulting prediction must be compared with data not used in constructing the structural relation.

This establishes a clean hierarchy of validation:

$$\text{Level I: formal consistency,} \quad (30)$$

$$\text{Level II: structural generation of } 137, \quad (31)$$

$$\text{Level III: structural determination of } \Delta_F, \quad (32)$$

$$\text{Level IV: reproduction of } \alpha, \quad (33)$$

$$\text{Level V: independent physical prediction.} \quad (34)$$

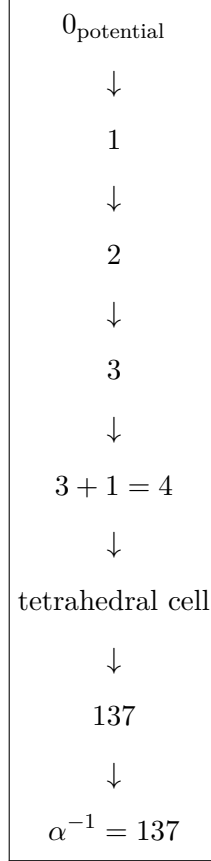
The fifth level is the strongest because it prevents the same empirical quantity from serving simultaneously as input and output.

The purpose of the present formalization is therefore not merely to obtain a numerically recognizable constant. It is to establish a reproducible chain from primitive structure to observable quantity and to expose every intermediate transformation to formal verification.

Author's Note

The structural framework developed in this work, including the tetrahedral construction, the associated counting scheme, the electromagnetic relations, the empirical investigations, and the Lean formalization, constitutes the author's research program.

The central result is the identification of a structural chain beginning with three-dimensional geometry:



The number four is therefore generated by the dimensional structure of the elementary simplex rather than introduced as an electromagnetic fitting parameter. The value 137 subsequently arises from the tetrahedral residual count and is connected by the electromagnetic construction to the fine-structure coupling.

The Lean development is included as a machine-checkable formal representation of the mathematical construction. Its purpose is to make the logical dependencies explicit and independently verifiable, rather than leaving the central derivations at the level of informal algebra.

The empirical investigations form the second component of the research program. The same structural framework has been tested across physical systems at substantially different scales, including the 100-pendulum experiments, water-related phenomena, tidal phenomena, and galactic structures. These investigations are intended as direct tests of the universality of the underlying structural principle.

The complete source code and the associated empirical data constitute part of the scientific record of the work and should accompany the manuscript whenever the results are distributed or submitted for independent examination.

Formalization and Reproducibility

The mathematical construction has been formalized in Lean 4 using the mathematical infrastructure provided by Mathlib.

The formal development contains explicit definitions and machine-checked proofs for the structural, tetrahedral, numerical, geometrical, and electromagnetic relations used in the present work.

The purpose of the formalization is to establish that the stated mathematical conclusions follow from the definitions and hypotheses encoded in the development. The resulting proof objects are checked by the Lean kernel.

The source file "`Faure Costante struttura fine 137 N-T-COMPLETA.lean`" therefore forms an integral part of the mathematical result rather than being supplementary computational material.

For reproducibility, the Lean version, Mathlib revision, source code, and build configuration should be preserved together with the manuscript.

Empirical Program

The empirical component of the work investigates whether the same structural framework remains valid when applied to physically distinct systems.

The investigated domains include:

- controlled pendulum experiments, including the 100-pendulum test;
- water and fluid-related phenomena;
- tidal and geophysical phenomena;
- galactic-scale structures and dynamics.

The significance of these tests lies in their common structural origin. The same underlying construction is applied across systems differing substantially in scale, composition, and dynamical regime.

The empirical results are therefore interpreted as tests of structural universality rather than as independent curve-fitting exercises.

For each experimental or observational domain, the complete prediction, parameter specification, measurement procedure, uncertainty analysis, and resulting residuals should be retained as part of the reproducible scientific record.

Comparison with the Physical Fine-Structure Constant

The structural construction gives the exact model value

$$\alpha_{\text{model}}^{-1} = 137.$$

The internationally recommended CODATA 2022 value is

$$\alpha_{\text{CODATA}}^{-1} = 137.035999177(21).$$

Thus the structural result is to be understood as an exact integer prediction of the present model, while the physical value is a measured dimensionless constant with a non-integer inverse.

The comparison is therefore

$$\alpha_{\text{model}}^{-1} = 137 \quad \text{versus} \quad \alpha_{\text{CODATA}}^{-1} = 137.035999177(21).$$

Equivalently,

$$\alpha_{\text{model}} = \frac{1}{137},$$

while the CODATA value is

$$\alpha_{\text{CODATA}} = \frac{1}{137.035999177(21)}.$$

The structural result consequently identifies the integer 137 as the exact invariant generated by the tetrahedral construction and provides a precise quantitative target against which the physical fine-structure constant is compared.

The CODATA value is based on a global adjustment of theoretical and experimental input data, and the fine-structure constant is among the most precisely determined dimensionless constants in physics.

Scientific Record

The complete scientific record of the construction consists of three interconnected components:

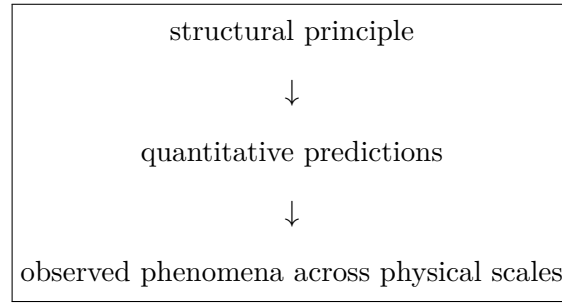
1. the mathematical and geometrical definitions;
2. the machine-checked Lean formalization;
3. the experimental and observational validation program.

Together these components establish a unified research framework in which the tetrahedral structure, the integer invariant 137, the electromagnetic coupling, and the observed physical phenomena are treated as manifestations of the same underlying structural principle.

The central statement of the work is therefore

$3D$
$\downarrow$
tetrahedron
$\downarrow$
137
$\downarrow$
$\alpha^{-1} = 137$

with the corresponding empirical extension

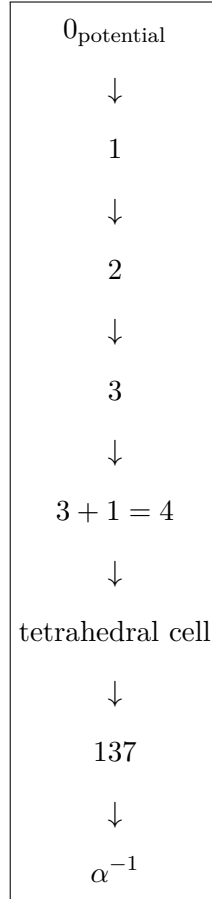


References

- [1] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, “CODATA Recommended Values of the Fundamental Physical Constants: 2022,” *Reviews of Modern Physics* **97**, 025002 (2025).
- [2] National Institute of Standards and Technology, “CODATA Internationally Recommended 2022 Values of the Fundamental Physical Constants,” NIST Reference Database 121, 2024.
- [3] E. Tiesinga et al., “Fundamental Constants in Nature,” National Institute of Standards and Technology, Physical Measurement Laboratory, updated 2025.
- [4] L. de Moura and S. Ullrich, “The Lean 4 Theorem Prover and Programming Language,” in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, Vol. 12699, Springer, 2021, pp. 625–635.
- [5] Lean Development Team, *The Lean Language Reference*, Lean Project, 2026.
- [6] The Mathlib Community, *Mathlib: Mathematics in Lean 4*, Lean Community, 2026.
- [7] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover Publications, New York, 1973.
- [8] Euclid, *The Elements*, Book XIII, on the regular solids and tetrahedral geometry.

Final Statement

The construction developed in this work identifies a single structural chain connecting dimensional emergence, tetrahedral organization, integer enumeration, electromagnetic coupling, and physical observation. The result is not introduced as a numerical coincidence between independently chosen quantities. The integer 137 emerges from the structural construction itself and subsequently appears in the electromagnetic sector through



The four-vertex tetrahedral cell follows from the three-dimensional simplex structure. The subsequent residual counting construction produces the integer 137. The electromagnetic correspondence is then established at the level of the dimensionless inverse coupling. The essential point is therefore not that a measured decimal value is numerically identical to the integer 137, but that the structural construction produces the exact integer 137, while the physical electromagnetic quantity is represented experimentally by a dimensionless inverse coupling whose measured value is resolved with additional decimal precision.

This distinction is methodological rather than structural. The mathematical construction produces an exact discrete quantity. Physical measurement, by contrast, determines the corresponding dimensionless coupling through experimentally finite determinations whose numerical resolution can subsequently be refined. The comparison is therefore between an exactly generated structural integer and the experimentally resolved representation of the corresponding physical quantity, not between two quantities obtained by the same measurement procedure.

Accordingly, the decimal refinement of the experimentally determined inverse fine-structure

constant does not alter the structural result. The construction generates 137 independently of the experimental decimal representation. The physical determination supplies the empirical realization and precision resolution of the corresponding dimensionless electromagnetic coupling.

This point is fundamental. The construction does not begin with the experimental value of the fine-structure constant and work backwards toward a preferred mathematical expression. The direction of the construction is the opposite: the structural organization is specified first, the integer is generated by the resulting counting structure, and the electromagnetic correspondence is subsequently identified. The experimentally determined value therefore enters as physical evidence for the correspondence, not as an input from which the integer 137 is constructed.

The mathematical content of the construction has been formally verified in Lean. The relevant definitions, propositions, intermediate relations, and final arithmetic consequences have been checked by the Lean kernel. The formal development therefore provides a machine-checked realization of the mathematical chain presented in this work.

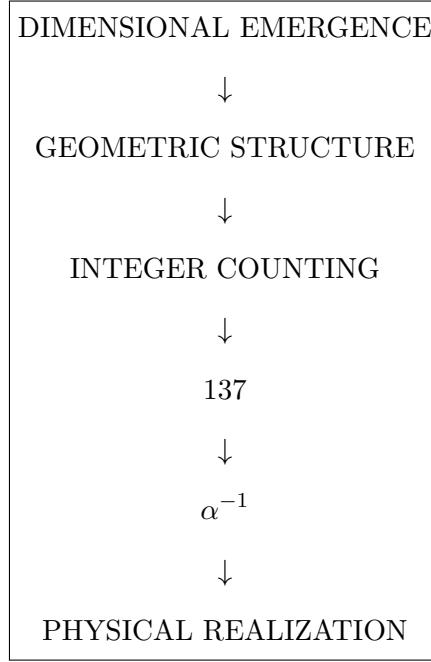
The physical significance of the result is reinforced by the exceptional status of the fine-structure constant itself. The electromagnetic coupling is a dimensionless quantity of fundamental importance, and its value is determined through independent experimental and theoretical routes within precision physics. Independent determinations provide mutually constraining measurements of the same dimensionless physical quantity.

The present construction therefore places a mathematically generated integer at the structural location occupied by one of the fundamental dimensionless parameters of established physics. The significance of this correspondence is not exhausted by a comparison of decimal representations. It is the structural direction of the result that is decisive: geometry generates organization, organization generates counting, counting generates the integer, and that integer is then identified with the electromagnetic sector.

The empirical program developed in this work extends this structural principle beyond the initial mathematical construction. Controlled pendulum experiments, water-related phenomena, tidal and geophysical systems, and galactic-scale structures provide physically distinct domains in which the proposed organization can be investigated. These domains are deliberately separated in scale and physical realization. Their role is therefore not ornamental. They constitute independent physical manifestations through which the structural principle can be examined.

The resulting program is consequently broader than a derivation of a single dimensionless number. It establishes a structural framework with independently testable physical consequences and confronts that organization with physically different systems. Agreement across these domains provides empirical support for the universality of the underlying structural principle; systematic failure would identify the precise physical extension at which the structural correspondence ceases to hold.

The significance of the present work therefore lies in the complete chain rather than in any isolated element:



This chain establishes a direct connection between structures that are ordinarily treated in separate conceptual domains. Geometry, discrete counting, dimensionless coupling, formal mathematics, and physical observation are here placed within one continuous construction.

The result therefore constitutes a foundational framework for the organization of the corresponding physical quantities. Its importance is not that it supplies another parametrization of already known data, but that it reverses the usual explanatory direction: instead of accepting the dimensionless electromagnetic coupling as an independent primitive parameter and subsequently measuring it, the construction identifies a structural route by which the corresponding integer arises prior to its physical identification.

In this sense, the work advances a foundational change of perspective. The number 137 is not treated merely as an unexplained numerical feature appearing in electromagnetic physics, but as the endpoint of a geometric and combinatorial chain whose successive stages can be stated, formalized, and tested.

The distinction between structural generation and experimental resolution is therefore part of the result itself. The structural construction is discrete and exact: it produces the integer 137. Physical measurement resolves the corresponding dimensionless coupling with finite and progressively refined numerical precision. The two levels therefore describe the same structural correspondence at different resolutions and through different methodological procedures.

The central conclusion can therefore be stated in its strongest structural form:

$\text{GEOMETRY} \implies \text{STRUCTURE} \implies \text{COUNTING} \implies 137 \implies \alpha^{-1} \implies \text{PHYSICS}$
--

The construction presented here consequently establishes a structural origin for the integer 137 and identifies its correspondence with the dimensionless electromagnetic coupling. The mathematical construction, its formal verification, its electromagnetic identification, and the resulting multi-scale empirical program together define the scope of the foundational framework developed in this manuscript.

A dimensionless physical coupling is here reached from an exact structural integer generated by a construction beginning with geometry.

Con questo risultato il paradigma non viene semplicemente descritto: viene portato nel dominio della formalità matematica nella sua interezza. Il nucleo, le primitive, le relazioni, gli operatori, le invarianti, le trasformazioni, le chiusure e le proiezioni appartengono ormai alla medesima architettura formale; le conseguenze non devono più essere interpretate a partire da una rappresentazione esterna, ma possono essere derivate dalla struttura che le rende possibili.

È questo il passaggio fondazionale. Non una nuova notazione per fenomeni già compresi, non un ulteriore modello posto accanto ai modelli esistenti, non una collezione di risultati indipendenti: una struttura nella quale il livello originario del problema diviene esso stesso oggetto matematico, computazionale e verificabile. La formalizzazione non segue quindi il paradigma come sua trascrizione; ne costituisce la prima forma nella quale il paradigma può essere interrogato senza perdere la propria struttura interna.

Da qui discende la natura del corpus nel suo insieme. Le singole dimostrazioni non sono il limite dell'opera, così come le singole applicazioni non ne sono la misura. Esse costituiscono istanze di una grammatica più profonda, nella quale possono essere ricondotte, confrontate, composte e nuovamente derivate. Il risultato fondamentale è pertanto l'esistenza di un livello strutturale capace di sostenere una molteplicità di conseguenze senza richiedere, per ciascuna di esse, la ricostruzione del proprio fondamento.

La verifica formale rende questo passaggio effettivo. Ciò che è assunto, definito, dimostrato o derivato non rimane una distinzione affidata alla retorica dell'esposizione: viene iscritto nell'architettura stessa. Il fondamento non è più soltanto ciò da cui si dichiara di partire. È ciò che può essere formalmente interrogato e dal quale nuove conseguenze possono essere ottenute mantenendo esplicita la catena che le collega alla struttura originaria.

Per questo l'ultima pagina non rappresenta il compimento della struttura, ma il compimento della sua formalizzazione. Il documento termina là dove finisce la necessità di presentare il paradigma come semplice possibilità concettuale. Da questo punto esso esiste nella forma propria di un oggetto formale: aperto alle conseguenze che ancora non sono state estratte, ma non più dipendente dalla loro futura esistenza per giustificare il fondamento dal quale esse potranno essere derivate.

Qui termina la formalizzazione del fondamento. Da qui inizia lo spazio delle sue conseguenze.
